

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		97204	
Total Threats Mitigated	24	Total Packets Analyzed		10253309629	
High Priority	10	Malicious Packets		10229722003	
Low Priority	14	True Negatives Passed		86998	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		0	
False Positives	88	UDP Flood		15	
FP Rate	0.10%	Uncertain		9	
IF Precision	99.14%				
IF Recall	95.07%	Action		Count	
IF F1-Score	97.06%	Quarantined		5	
IF Accuracy	99.37%	Time Ban		3	
		Blackhole		3	
		Released		6	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.14%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.07%	Share of actual attacks successfully flagged
F1-Score	97.06%	Balanced measure combining Precision and Recall
Accuracy	99.37%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.10%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.93%	Actual attacks that went undetected
True Positive Rate (TPR)	95.07%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.90%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 10093	FN 523
Actual: Normal	FP 88	TN 86475

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	10093

2c. Response Latency

Metric	Value	Description
Detection Time	326.3 ms	Average time taken to detect an attack
Mitigation Response Time	6.8 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.15%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.97%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.89%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.6	1	7	2026-09-14 11:01:14	2026-09-14 11:01:24
10.0.0.20	10	3	2026-09-14 10:44:29	2026-09-14 11:06:11
10.0.0.21	9	2	2026-09-14 10:44:30	2026-09-14 11:06:11
10.0.0.22	9	2	2026-09-14 10:44:31	2026-09-14 11:06:13
10.0.0.8	2	2	2026-09-14 10:51:11	2026-09-14 10:52:47
10.0.0.2	2	1	2026-09-14 10:51:10	2026-09-14 10:51:53
10.0.0.5	2	1	2026-09-14 10:51:10	2026-09-14 10:51:54
10.0.0.10	1	1	2026-09-14 11:01:24	2026-09-14 11:01:34

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-14 10:44:29	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 10:44:30	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 10:44:31	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-14 10:44:40	10.0.0.20	DDoS	UDP Flood	76.0%	High	Time Ban (1m)
2026-09-14 10:44:40	10.0.0.20	DDoS	UDP Flood	78.9%	Critical	Time Ban
2026-09-14 10:44:41	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (1m)
2026-09-14 10:44:41	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban
2026-09-14 10:44:42	10.0.0.22	DDoS	UDP Flood	78.6%	High	Time Ban (1m)
2026-09-14 10:44:42	10.0.0.22	DDoS	UDP Flood	78.6%	High	Time Ban
2026-09-14 10:45:40	10.0.0.20	DDoS	UDP Flood	83.3%	High	Released
2026-09-14 10:45:41	10.0.0.21	DDoS	UDP Flood	78.1%	High	Time Ban (2m)
2026-09-14 10:45:42	10.0.0.22	DDoS	UDP Flood	78.6%	High	Time Ban (2m)
2026-09-14 10:45:51	10.0.0.20	DDoS	UDP Flood	83.3%	Critical	Time Ban (2m)
2026-09-14 10:51:11	10.0.0.8	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-14 10:51:20	10.0.0.2	DDoS	Uncertain	58.9%	Low	Released
2026-09-14 10:51:20	10.0.0.5	DDoS	Uncertain	58.9%	Low	Released
2026-09-14 10:52:11	10.0.0.8	DDoS	UDP Flood	92.4%	High	Released
2026-09-14 10:52:36	10.0.0.8	DDoS	Uncertain	69.2%	Medium	Quarantined
2026-09-14 10:56:54	10.0.0.20	DDoS	UDP Flood	80.8%	Critical	Blackhole
2026-09-14 11:01:07	10.0.0.21	DDoS	UDP Flood	77.9%	Critical	Blackhole
2026-09-14 11:01:10	10.0.0.22	DDoS	UDP Flood	77.4%	Critical	Blackhole
2026-09-14 11:01:14	10.0.0.6	DDoS	Uncertain	68.7%	Medium	Quarantined
2026-09-14 11:01:24	10.0.0.6	DDoS	Uncertain	68.7%	Medium	Released
2026-09-14 11:01:34	10.0.0.10	DDoS	Uncertain	68.7%	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6288	75.0%	Critical	Phase 3	1	1s	2026-09-14 11:06:13	Escalated to Blackhole
10.0.0.21	UDP Flood	0.6448	75.8%	Critical	Phase 3	1	0s	2026-09-14 11:06:11	Escalated to Blackhole
10.0.0.20	UDP Flood	0.6261	76.0%	Critical	Phase 3	1	0s	2026-09-14 11:06:11	Escalated to Blackhole
10.0.0.10	Uncertain	0.5708	68.7%	Low	Phase 1	1	10s	2026-09-14 11:01:34	Attack Stopped
10.0.0.6	Uncertain	0.6345	68.7%	Medium	Phase 1	7	10s	2026-09-14 11:01:24	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6411	77.4%	Critical	Phase 3	2	12s	2026-09-14 11:01:10	Escalated to Blackhole
10.0.0.22	UDP Flood	0.6402	77.4%	Critical	Phase 2	1	11s	2026-09-14 11:01:09	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6295	77.9%	Critical	Phase 2	1	10s	2026-09-14 11:01:07	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6295	77.9%	Critical	Phase 3	2	10s	2026-09-14 11:01:07	Escalated to Blackhole
10.0.0.20	UDP Flood	0.6466	75.4%	Critical	Phase 3	1	1s	2026-09-14 11:00:58	Escalated to Blackhole
10.0.0.22	UDP Flood	0.6346	79.3%	Critical	Phase 2	2	1m 11s	2026-09-14 10:56:55	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6518	80.8%	Critical	Phase 2	2	1m 11s	2026-09-14 10:56:54	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6303	77.6%	Critical	Phase 2	2	1m 11s	2026-09-14 10:56:54	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6467	80.8%	Critical	Phase 3	3	1m 11s	2026-09-14 10:56:54	Escalated to Blackhole
10.0.0.22	UDP Flood	0.6330	76.7%	Critical	Phase 2	1	11s	2026-09-14 10:55:55	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	75.9%	Critical	Phase 2	1	11s	2026-09-14 10:55:54	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6297	75.7%	Critical	Phase 2	1	11s	2026-09-14 10:55:54	Attack persisted - escalating to time ban
10.0.0.8	Uncertain	0.5893	69.2%	Low	Phase 1	2	11s	2026-09-14 10:52:47	Attack Stopped
10.0.0.8	UDP Flood	0.5623	92.4%	High	Phase 2	1	1m 0s	2026-09-14 10:52:11	Ban Expired
10.0.0.5	Uncertain	0.6075	58.9%	Low	Phase 1	1	10s	2026-09-14 10:51:54	Attack Stopped
10.0.0.2	Uncertain	0.6027	58.9%	Low	Phase 1	1	10s	2026-09-14 10:51:53	Attack Stopped
10.0.0.22	UDP Flood	0.6192	86.3%	High	Phase 2	2	1m 10s	2026-09-14 10:51:40	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6400	79.2%	High	Phase 2	2	1m 10s	2026-09-14 10:51:39	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6390	80.0%	High	Phase 2	2	1m 10s	2026-09-14 10:51:39	Attack persisted - escalating to time ban
10.0.0.2	Uncertain	0.6000	58.9%	Low	Phase 1	1	10s	2026-09-14 10:51:20	Attack Stopped
10.0.0.5	Uncertain	0.6062	58.9%	Low	Phase 1	1	10s	2026-09-14 10:51:20	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6266	86.3%	Critical	Phase 2	1	10s	2026-09-14 10:50:40	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	75.6%	Critical	Phase 2	1	10s	2026-09-14 10:50:39	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6458	77.9%	Critical	Phase 2	1	10s	2026-09-14 10:50:39	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6353	83.3%	Critical	Phase 2	3	11s	2026-09-14 10:45:51	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6185	78.6%	High	Phase 2	2	1m 11s	2026-09-14 10:45:42	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6111	78.1%	High	Phase 2	2	1m 11s	2026-09-14 10:45:41	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6105	83.3%	High	Phase 2	1	1m 11s	2026-09-14 10:45:40	Ban Expired
10.0.0.22	UDP Flood	0.6250	78.6%	High	Phase 2	1	11s	2026-09-14 10:44:42	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6189	78.1%	High	Phase 2	1	11s	2026-09-14 10:44:41	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	76.0%	High	Phase 2	1	11s	2026-09-14 10:44:40	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE